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Hydrogen-rich fuel combustion in a swirling flame: CFD-modeling with experimental verification

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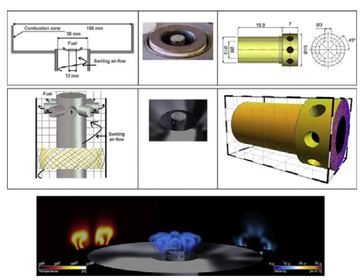
HIGHLIGHTS

- CFD-modeling and experiment of synthetic fuel combustion.
- Increase in molar fraction of hydrogen leads to increase in combustion temperature.
- Emission characteristics of synthetic fuel and pure CH_4, H_2, CO were compared.
- NO_x emission is 88.9 ppm to 93.1 ppm for synfuel with 0.4 and 0.8 H2 mole fraction.
- Minimum NO_x emission for synthetic fuel combustion.

GRAPHICAL ABSTRACT



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ABSTRACT

The CFD-modeling of synthetic (hydrogen-rich) fuel combustion in a swirling flame was performed. To simulate the turbulent flames with rigorous turbulence chemistry interaction, a PDF combustion model with detailed chemical kinetics was used. The results of numerical simulations were compared with experimental data. The study was carried out for synthetic fuel, which consists of methane, hydrogen, carbon monoxide, carbon dioxide, and nitrogen. The composition of the synthetic fuel was varied due to a change in the molar fraction of hydrogen from 0.4 to 0.8, and due to a decrease in the molar fraction of CO from 0.44 to 0.04. The flame contours, temperature, and NOx emissions were compared between numerical and experimental data. A visual comparison of OH^- contours showed good convergence. The discrepancy between the results of experiment and CFD-modeling is about 3% for temperature and less than 7% for NO_x emission. The NOx emissions for synthetic fuel and pure methane, hydrogen, carbon monoxide are compared. For all investigated the synfuel compositions, the NO_x emission was from 88.9 ppm to 93.1 ppm for synfuel with 0.4 and 0.8 hydrogen mole fraction, respectively.

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Introduction

Traditional fossil fuels, such as oil and natural gas, are still the main source of thermal energy for various fuel-consuming plants. In addition, according to the International Energy Agency's (IEA) report "Golden Rules for a Golden Age of Gas" [1,2], natural gas will be the main fuel throughout the 21st century due to good environmental and economic characteristics. However, despite the widespread use of natural gas, the focus of engineers and scientists in various countries is aimed at finding alternative types of gaseous fuels. One such alternative type of gaseous fuel is synthetic fuel. Synthetic fuel (or synfuel) – hydrocarbon fuel that differs from traditional fuel in the production process. Synthetic fuel is produced by processing the original hydrocarbon fuel [3–5]. The composition of the synthetic fuel can vary significantly depending on the production conditions and the type of initial hydrocarbons. However, hydrogen and carbon monoxide is the main combustible components of synthetic fuel.

Synthetic fuel as the main source of thermal energy is used in fuel-consuming plants with thermochemical waste-heat recuperation [6,7]. In such plants, the generation of thermal energy is divided into two stages. The first is the production of synthetic fuel due to an endothermic fuel reforming process. The second stage is the combustion of synthesized fuel that mainly consists of hydrogen and carbon monoxide. The investigation of thermochemical recuperation systems has been widely discussed in the scientific literature for GTU [8], ICE [9], industrial furnaces [10].

Moreover, one of the main source of thermal syngas is coal and biomass gasification processes. Advanced power systems (APS) that are projected to get high efficiency and low NO_x emissions, for example the program "Vision 21" [11], rely on syngas as a key intermediate energy source. In such APS, coal or hydrocarbon fuel are reformed to syngas (main combustible components are carbon monoxide (CO) and hydrogen (H_2)) via gasification process, partial oxidation, and fuel reforming. The targets outlined for Vision 21 powerplants are 75% thermal efficiency for natural gas fueled plants on a LHV basis and 60% for coal fueled plants on a HHV basis while producing electricity, without CO_2 capture and sequestration or co-production of any transportation fuels.

An important task when using synthetic fuel is to determine the characteristics of synfuel combustion, the structure, and properties of the flame. A distinctive feature of calculating the combustion characteristics of synthetic fuel is the fact that the composition of synthetic fuel can vary significantly depending on the conditions for obtaining such synthetic fuel [12]. In addition, the variety of burner designs also requires universal calculation methods. One way to determine these characteristics is CFD-modeling. Numerical modeling of various physicochemical processes in engineering and technology using specialized software in recent years is increasingly used in the work of engineers and scientists from different countries. According to the NASA "Vision 2030" report [13], by 2030 about 80% of the computing power of all computers on Earth will be spent on solving problems of numerical modeling of various processes and objects.

There are a significant number of publications dealing with CFD modeling of gaseous fuel combustion [14–16]. In the literature, there are articles devoted to CFD-modeling the combustion of methane [17], hydrogen [18], synthesis gas [19]. A wide range of currently solved problems allows us to state that with the help of special software products, such as ANSYS [20,21], Comsol [22], OpenFOAM [23–26], it is possible to solve a lot of the known scientific and technical problems related to the fuel combustion.

Numerical simulation of combustion gives a good correlation of the results with experimental data. Gomes et al. presented the results of the application of the CFD simulation to the evaluation of natural gas replacement by syngas in burners of the ceramic sector [27]. The authors compared the results of CFD-modeling with the well-documented experimental results of NG combustion performed in the Burner Engineering Research Laboratory (BERL) [28]. The numerical results were in a good correlation with experimental data. Also, Jozwiak with co-worker [29] published the results of numerical and experimental investigation of the combustion of syngas and natural gas mixture for the industrial furnaces. Such type of fuel (a mixture of syngas and natural gas) is widely used in the industrial furnaces with the thermochemical waste-heat recuperation systems. In the author's work with the help of CFD-modeling, the accurate results on combustion and emission characteristics were obtained. Yang et al. [30] developed a CFD-model of a micro tube combustor with converging-diverging channel and based on the developed model their numerically investigated the thermal performance and compared with that of the micro combustor with cylindrical channel. Mezaine ad co-workers [16] presented the results of a numerical investigation of the effect of natural gas-hydrogen blended fuel on the combustion performance of a Rich/Quench/Lean (RQL) can combustor. In their model, the a reduced GRIMECH 3.0 mechanism was adopted as the reaction scheme of the combustion process for the developed CFD-model. Therefore, a variety of publications on the CFD-modeling of fuel combustion shows the wide possibilities of using this instrument to determine the combustion characteristics of various fuels.

With the aim of the determination of combustion and emission characteristics for synfuel, the CFD-model of premixed combustion in a swirling flame was developed. The experimental investigation of synfuel combustion was performed to obtain the accurate results for verification of the numerical results. As a result of comparing experimental data with the numerical results, the scientific justification and validation of the physicomathematical approaches for CFD-modeling of combustion of multicomponent hydrogen-containing gas mixtures were presented.

Experimental and numerical setup

Geometry

The numerical and experimental investigation of synfuel combustion was performed in a swirl-burner where air before combustion was swirled. The geometry of the burner for the CFD-model has the same dimensions as a real burner. The

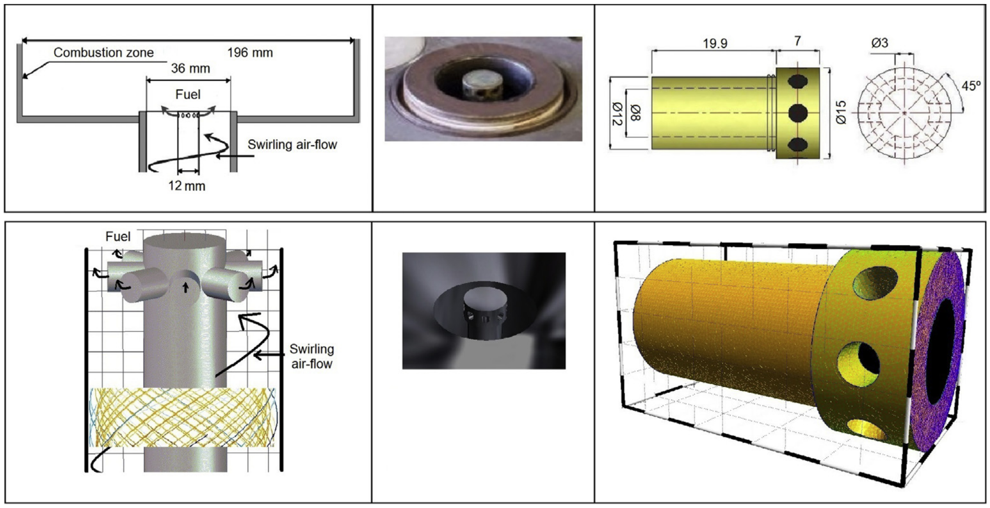

Fig. 1 — Flow scheme in the combustion process; 3D-model (bottom) and photo of an experimental burner (top) [39].

scheme of flows in the burner and comparison of the numerical and real burner is presented in Fig. 1.

The main geometric dimensions of the burner are shown in Fig. 1. The burner consists of 8 nozzles through which premixed synthetic fuel is supplied. In addition, swirling airflow is supplied through the main nozzle of the burner assembly. A general view of the computational domain and the computational mesh is shown in Fig. 2.

A gas burner is installed in a sealed combustion chamber. The maximum thermal power of the burner is 20,375 W. The combustion products after the combustion chamber are sampled by a water-cooled probe. Further, the combustion products are cooled to a temperature of approximately 500 °C. The composition of the combustion products is measured using a DAG-16 (дАг-16 in Russian) gas analyzer. This gas analyzer is measured NO_x emission as one value for all nitrogen oxides (NO, N_2, NO_2). Temperature is measured using a TI 315E infrared pyrometer. The synfuel was obtained by mixing separate gases from the cylinders. The gases (H_2, CO, CO_2, CH_4) in the cylinders have high purity (N40 class). Gas consumption is measured using an electronic balance with a graduation scale of 0.001 g.

The mesh generation was performed in ANSYS Mesh. The total number of mesh elements is about 7 million. To choose this number of mesh elements, the mesh independence study was performed. Five meshes were checked: 0.5, 1.0, 2.5, 5.0, and 7.0 million. The mesh with 7-million elements is maximum for the used PC for CFD-modeling. The mesh structure was the same for all meshes. The combustion temperature was chosen as a control parameter. The results of CFD-modeling were compared with the experimental data. The minimal discrepancy between the experimental data and the results of CFD-modeling was obtained for a mesh of 7 million elements. The mesh structure is shown in Fig. 2. The unit elements are tetrahedrons of various sizes. For the flowing region near the walls, for the correct modeling of the boundary layer, mesh thickening was used. The y^+ was estimated for the computational mesh. The mesh was adapted to the dimensionless distance from the first grid node to the wall (y^+). To adapt the mesh for y^+, the procedure adapt in Fluent solver was used. Adaptation is made in such a way that the condition is satisfied - $y^+ < 50$. For the main part of the flow area, the parameter y^+ does not exceed 25, only in the areas of contact of the particle surfaces with each other and with the pipe wall $25 < y^+ < 50$. This is achieved by introducing a constraint in the calculations of y^+ in such a way that the following condition is satisfied:

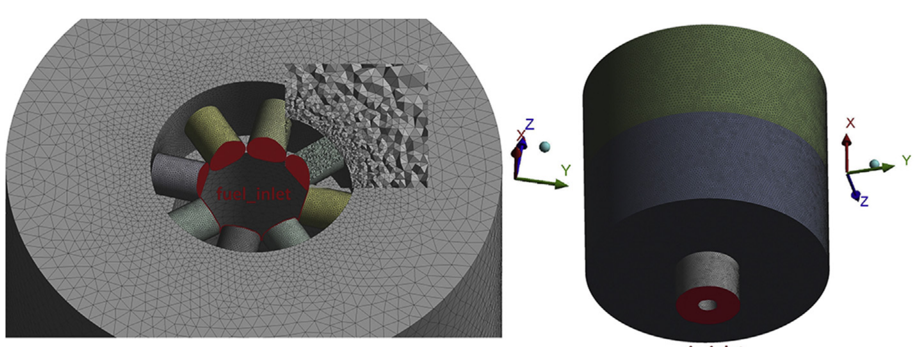

Fig. 2 — Computational domain and mesh structure for developed CFD-model.

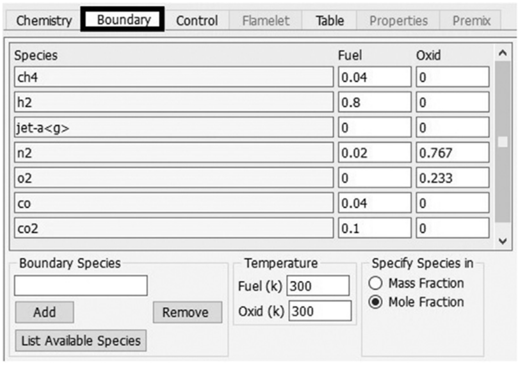

Fig. 3 – The synfuel and oxidizer composition, and initial temperature of fuel and oxidizer.

$$\overline{y}^{+} = MAX(y^{+}; y^{+}_{\lim}) \tag{1}$$

where y^{+}_{lim} is maximum allowed value for each element of the mesh.

The synfuel compositions were set in a species transport module as presented in Fig. 3. The composition of synfuel is close to the synfuel composition obtained in the thermochemical waste-heat recuperation systems [31–33]. The temperature of fuel and air was set at 300 K.

Governing equations

In this study, synfuel burns in a swirling air-flow at various air-to-fuel ratio (fuel/air equivalence ratio). For numerical calculation, the continuity, momentum, and energy equations with the chemical combustion kinetics were solved. The Ansys Fluent (full academic version) was used for this task. The CFD-modeling was performed for transient (unsteady) flow. The following assumptions were used for the model:

- surface chemical oxidation of the burner components do not occur;
- work by viscous forces and by pressure is not done;
- no the flux of energy due to a mass concentration gradient (no Duflor effects [34]).

The continuity equation for unsteady flow can be written as follows:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0 \tag{2}$$

The momentum equation in an inertial (non-accelerating) reference frame is described by follow equations [35]:

$$\frac{\partial}{\partial t}(\rho \vec{v}) + \nabla \cdot (\rho \vec{v}\,\vec{v}) = -\nabla \cdot p + \nabla \cdot (\overline{\overline{\tau}}) + \rho \vec{g} + \vec{F} \tag{3}$$

where p is the static pressure; $\rho \vec{g}$ and $\vec{F}$ are the gravitational body force and external body forces, respectively. The stress tensor $\overline{\overline{\tau}}$ is given by follows equation:

$$\overline{\overline{\tau}} = \mu\left[\left(\nabla \vec{v} + \vec{v}^{T}\right) - \frac{2}{3}\nabla \cdot \vec{v} I\right] \tag{4}$$

where μ is the molecular viscosity, I is the unit tensor, and the second term on the right hand side is the effect of volume dilation.

The processes in the packed beds are taking place at the wide ranges of velocity. The flow around the particles may have a vortex after the particle. Therefore in the CFD-model, the RNG k-ε model is chosen.

$$\frac{\partial}{\partial x_i}(\rho k v_i) = \frac{\partial}{\partial x_j}\left(\alpha_k \mu_{eff} \frac{\partial k}{\partial x_j}\right) + G_k + G_b - \rho\varepsilon - Y_M + S_k \tag{5}$$

and

$$\frac{\partial}{\partial x_i}(\rho \varepsilon v_i) = \frac{\partial}{\partial x_j}\left(\alpha_\varepsilon \mu_{eff} \frac{\partial \varepsilon}{\partial x_j}\right) + C_{1\varepsilon}\frac{\varepsilon}{k}(G_k + C_{3\varepsilon}G_b) - C_{2\varepsilon}\rho\frac{\varepsilon^2}{k} - R_\varepsilon + S_\varepsilon \tag{6}$$

In Eqs. (5) and (6) there are following terms:

- G_k – the generation of turbulence kinetic energy due to the mean velocity gradients;
- Y_M – the contribution of the fluctuating dilatation in compressible turbulence to the overall dissipation rate;
- G_b – the generation of turbulence kinetic energy due to buoyancy;
- α_k and α_ε – the inverse effective Prandtl numbers for k and ε, respectively;

The additional term R_ε in Eq. (6) equation can be defined as:

$$R_\varepsilon = \frac{C_\mu \rho \eta^3 (1 - \eta/\eta_0)}{1 + \beta\eta^3}\frac{\varepsilon^2}{k} \tag{7}$$

where $\eta \equiv Sk/\varepsilon$, $\eta_0 = 4.38$, $\beta = 0.012$.

The model constants $C_{1\varepsilon}$ and $C_{2\varepsilon}$ in for Eqs (5) and (6) used by default setup for Fluent solver of ANSYS [35]:

$$C_{1\varepsilon} = 1.42; \quad C_{2\varepsilon} = 1.68 \tag{8}$$

The energy conservation equation for a combustion zone can be determined as follows:

$$\nabla \cdot \vec{u}(p + \rho \cdot E_f) = \nabla \cdot \left[k_{eff} \cdot \nabla T - \left(\sum h_j \cdot \vec{J}_j\right) + \overline{\overline{\tau}} \cdot \vec{v}\right] + S_f^h \tag{9}$$

In this equation T is the combustion product temperature; ρ is the density; E_f is total fluid energy; $\vec{J}_j$ is the diffusion flux of j element; k_{eff} is the conductivity; h_j is the enthalpy of j element; S_f^h is the fluid enthalpy source term (include the source of energy due to chemical reaction).

The transport equation of combustion products is described by the following equations:

$$\nabla \cdot (\rho \cdot \vec{v} \cdot Y_j) = -\nabla \vec{J}_j + R_j \tag{10}$$

where R_j is chemical reaction rate of species j formation or decomposition; Y_j is the mass fraction of j element of combustion products; $\vec{J}_j$ is diffusion flux of j species.

Diffusion flux of j species can be written as:

$$\vec{J}_j = -\left(\rho \cdot D_{j,m} + \frac{\mu_t}{Sc_t}\nabla Y_j - D_{T,j}\frac{\nabla T}{T}\right) \tag{11}$$

where $D_{j,m}$ and $D_{T,j}$ are the mass diffusion coefficient and thermal diffusion coefficient for j species, respectively; Sc_t and

μ_t are the turbulent Schmidt number and turbulent viscosity, respectively.

To simulate synfuel combustion, data on the combustion kinetics of the main components of synfuel: methane, carbon monoxide, and hydrogen, was used. The composition PDF transport model is used for simulating finite-rate chemical kinetic effects in reacting flows. This model allows predicting with an appropriate chemical mechanism, kinetically-controlled species such as CO and NO_x, as well as flame extinction and ignition.

Due to the fact that the temperature in the flame front is more than 2000K, and triatomic components, such as steam and carbon dioxide, are located in the combustion products, the influence of radiant heat transfer on enthalpy losses in the gas stream will take place. Even though some authors believe [36] that the effect of radiant heat transfer during hydrogen-rich fuel combustion does not significantly affect the enthalpy loss, the model presented by the author uses the P-1 radiation model, which is a simplified version of the more general P–N model.

For the combustion zone the transport equation for the incident radiation for the gray, emitting, absorbing, and also scattering medium are defined as follows [37]:

$$\nabla \cdot (\Gamma \nabla G) + 4\pi \left(a \frac{\sigma T^4}{\pi} + E_p \right) - (a + a_p) G = 0 \tag{12}$$

In this equation Γ is the parameter of P-1 radiation model:

$$\Gamma = \frac{1}{(3(a + \sigma_s) - C\sigma_s)} \tag{13}$$

where σ_s is the scattering coefficient, a is the absorption coefficient of combustion products and wall, G is the incident radiation, C is the linear-anisotropic phase function coefficient.

In addition, in Eq. (12) the parameter E_p is the equivalent emission of the particles and a_p is the equivalent absorption coefficient depends on geometry characteristics. The mentioned values can be written as:

$$E_p = \lim_{V \to 0} \sum_{n=1}^{N} \varepsilon_{pn} A_{pn} \frac{\sigma T_{pn}^4}{\pi V} \tag{14}$$

and

$$a_p = \lim_{V \to 0} \sum_{n=1}^{N} \varepsilon_{pn} \frac{A_{pn}}{V} \tag{15}$$

In these equations the summation is over N particles in volume V.

In the calculations, the convergence criterion is set as 10^{-7} for the momentum equation and the continuity equation, 10^{-4} for the convergence of the energy equation. The equations of the CFD-model were solved in ANSYS-Fluent 18.2 (Academic Version) on a computing cluster with processors based on Intel Xeon E5-2640. The calculations were performed for the time step of 10^{-4} s and 10,000 steps (1 s of combustion). In the solution setup the solution parameters were retained as the default settings of ANSYS Fluent except [38]:

- Second Order Upwind for the Momentum;
- Least Squares Cell Based for the Gradient.

The under-relaxation factors for pressure, momentum, k, and ε were reduced from their default values to about 0.95. The under-relaxation factors for temperature was reduced to 0.8 because density is strongly coupled with temperature. The computation time for each calculation was on the order of about 8 h.

Results and discussion

Verification

A series of experiments on the combustion of synfuel with various compositions were performed to obtain the contours of the flame, to determine the temperature and NOx emissions. These experimental data were used to verify the CFD-model. Verification of the results of CFD-modeling is carried out in two stages: the first is a visual comparison of the structure of the flame; the second is a comparison of experimental temperature and emission with calculated values.

The comparison of the flame structure (OH^- contours) obtained via numerical simulation and experiment is shown in Fig. 4 [39]. The flame was obtained for synfuel composition presented in Fig. 3. The synfuel mainly consists of hydrogen. A

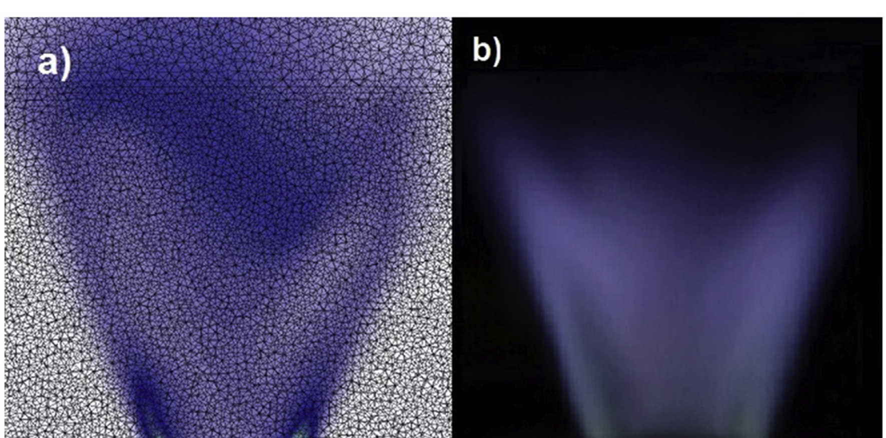

Fig. 4 – The comparison of the flame structure (OH^-) obtained in current study via numerical simulation (a) and experiment (b).

visual comparison shows good convergence of results. In the base of the flame, regions of increased OH^- concentration are visible both for the results of CFD modeling and experimental data. In addition, the angle of the flame disclosure for the numerical flame is 36°, for experiment 37°.

Fig. 5 shows the comparison of results obtained via CFD-modeling and experiment. This figure presents the temperature of combustion products and NO_x emission. The NO_x emission was measured with the help of Testo t350 XL gas analyzer. The screen of a display of gas analyzer is also presented in Fig. 5 for the following synfuel composition: $CH_4/H_2/N_2/CO/CO_2 = 0.04/0.6/0.02/0.24/0.1$. The temperature and NO_x emission was measured at the center of combustion zone (Fig. 1) at 300 mm above the fuel nozzle. Fig. 5 depicts the results obtained for various synfuel composition. The mole fraction of hydrogen was varied from 0.4 to 0.8. When the mole fraction of hydrogen was increased the mole fraction of carbon monoxide was decreased. The mole fractions of other synfuel components were the same as in Fig. 3. Fig. 5 shows a good correlation between the experimental and numerical results. One of the main advantages of a PDF-model to simulate synfuel combustion is a possibility to obtain an accurate value of NO_x emission. The root-mean-square error for temperature was about 3% and for NO_x emission about 6%. Root Mean Square Error (RMSE) measures how much error there is between the results of CFD-modeling and experiments. In other words, it compares a predicted value and an observed or known value. The smaller an RMSE value, the closer predicted and observed values are. The RMSE was calculated according to follows expression:

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(P_i - O_i)^2}{n}} \tag{16}$$

where P_i is the predicted results obtained via CFD-modeling; O_i is the observed results obtained via the experiments; n is number of values.

As can be seen from Fig. 5, an increase in the mole fraction of hydrogen leads to an increase in the combustion temperature. This is due to the fact that the heat combustion of hydrogen is higher than the heat combustion of carbon monoxide. In addition, with an increase in the mole fraction of hydrogen, the emission of NO_x increases. This is mainly due to the fact that with an increase in the combustion temperature, the reaction rate of the formation of nitrogen oxides increases.

The results in Figs. 4 and 5 show a good correlation between the experimental results and results obtained via CFD-modeling. Therefore, the numerical model can be used to wider investigate the practical interesting cases in synfuel combustion.

CFD-modeling results

The main advantage of CFD modeling is the visibility of the results and the possibility of their quick analysis. Fig. 6 shows the contours of the mass concentration of nitrogen oxides (a) and the contours of the temperature of the combustion products (b). It can be seen from Fig. 6 that the maximum concentration of nitrogen oxides is observed in the region of the maximum flame temperature. This fact agrees well with the classical concepts and mechanisms of the formation of nitrogen oxides during combustion. The contours in Fig. 6 was obtained for synfuel composition presented in Fig. 3.

The developed CFD-model allows determining not only combustion characteristics when the flow of reactants became steady. With the help of the CFD-model, the ignition of synfuel can be investigated. Fig. 7 shows the contours of temperature (left), OH^- (center), and velocity (right) at time step of 0.0001 s. Determination of the ignition characteristics for various synfuels at different initial temperatures is an important task for industrial furnaces with the

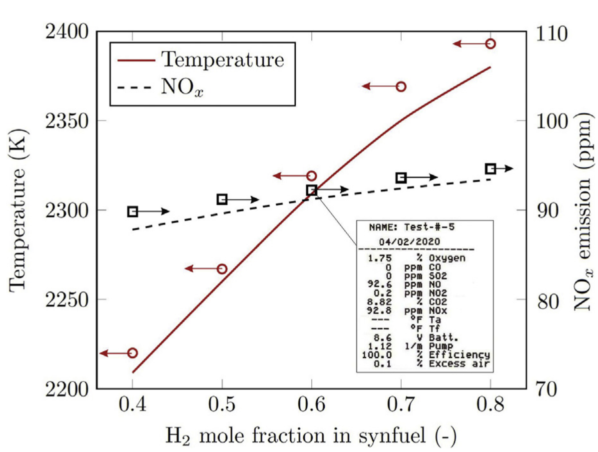

Fig. 5 – The comparison of temperature and NO_x emission obtained via experiment and CFD-modeling: hollow points – experimental results; lines – numerical results.

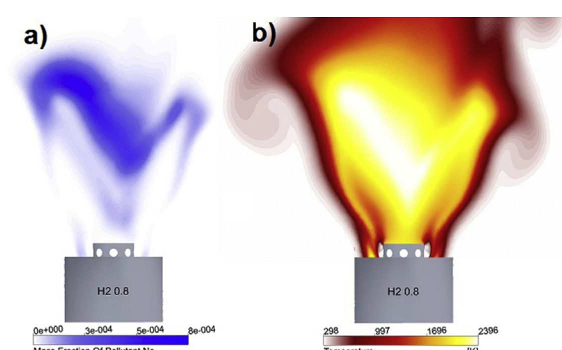

Fig. 6 – The comparison of temperature and NO_x emission obtained via experiment and CFD-modeling: hollow points – experimental results; lines – numerical results.

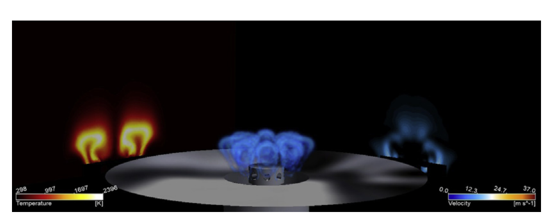

Fig. 7 – The comparison of temperature and NO_x emission obtained via experiment and CFD-modeling: hollow points – experimental results; lines – numerical results.

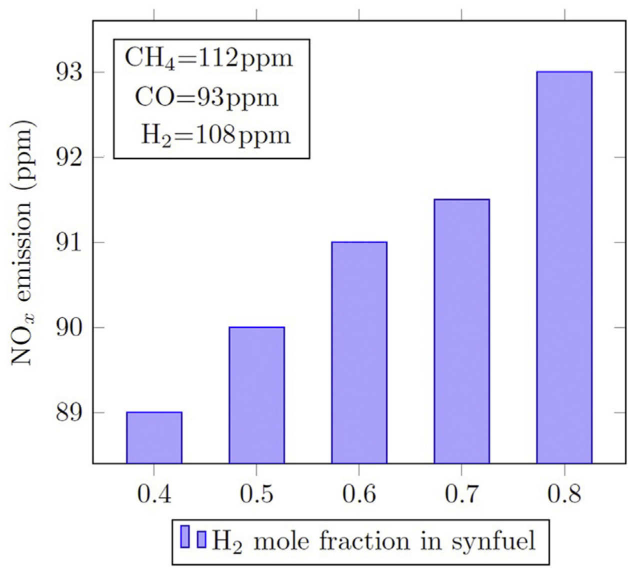

Fig. 8 – The comparison of NO_x emission calculated with the help of CFD-model for combustion of synfuel with various composition. The calculated NO_x emission for pure methane, carbon monoxide, and hydrogen combustion is given in a frame.

thermochemical heat recuperation systems because it makes it possible to determine the flame propagation velocity.

To assess the emission characteristics of synfuel combustion, the comparison of NO_x emission for combustion of synfuel of various compositions, methane, hydrogen, and carbon monoxide was performed. NO_x emission for the mentioned fuel was determined based on the developed CFD-model.

Fig. 8 presents the NOx emissions for the investigated fuels when at fuel/air equivalence ratio of 1.0. The results obtained via the developed CFD-model. The calculated NO_x emission for pure methane, carbon monoxide, and hydrogen combustion is given in a frame in Fig. 8. For all fuels, the burner output was the same ($\approx 20kW$). In the numerical experiment was determined that for synfuel combustion the NO_x emission is minimal for all investigated fuels. The maximum NO_x emission was observed for the methane combustion. Moreover, as it was established that with increasing in hydrogen mole fraction in synfuel, the NO_x emission is increasing. The lowest NO_x emission for synfuel can be explained by the fact that the swirl effect leads to a more complete mixing of the combustible components of the synfuel. Since the combustion heat of the synthetic fuel is less than the combustion heat of other investigated fuels, in order to maintain equal burner output, more fuel is supplied to the burner. In turn, greater fuel consumption provides more complete mixing. The low NO_x emissions at synfuel combustion show the big potential of synfuel for various fuel-consuming plants.

Conclusion

The results in this study illustrate the possibility of using modern software for numerical simulation to solve the problems of synthetic fuel combustion. A comparison of the obtained CFD results with experimental data shows good visual convergence of the flame contours, and a correlation between the results of CFD-modeling and experimental data for NO_x emission and temperature is obvious. The investigated synfuel combustion revealed a notable decrease in NO_x emissions comparing to traditional fuels such as methane, hydrogen, and carbon monoxide. For all investigated the synfuel compositions, the NO_x emission was from 88.9 ppm to 93.1 ppm for synfuel with 0.4 and 0.8 hydrogen mole fraction, respectively. The important result of this investigation is to show the way for determining the synthetic fuel composition that yields the lowest NO_x emission in synthetic fuel combustion.

Will numerical simulation in the future completely and partially replace a physical experiment in a combustion engineering largely depends on the accuracy and adequacy of the models of combustion, turbulence, radiation used. The presented work shows the possibility of successfully solving the problem of numerical simulation for the combustion of synthetic fuel. In this work, verification and the scientific substantiation of the models embedded in ANSYS Fluent were carried out as applied to the CFD simulation of synthetic fuel combustion.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at https://doi.org/10.1016/j.ijhydene.2020.05.113.

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